IMPACT OF GENDER INEQUALITY IN EDUCATION ON ECONOMIC DEVELOPMENT IN NIGERIA

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ABSTRACT

This study investigates the impact of gender inequality in education on economic development in Nigeria over the period 1981–2024. Using an Autoregressive Distributed Lag (ARDL) approach—selected for its flexibility with mixed I(0) and I(1) variables—the research models GDP per capita (PC) as a function of the Gender Parity Index (GPI), Labour Force Participation (LFP), Female Literacy Rate (FLR), Male Literacy Rate (MLR), Government Expenditure on Education (GEE), and Female Unemployment (UEMF). Data were sourced from the Central Bank of Nigeria (CBN) Statistical Bulletin, World Development Indicators (WDI), and the National Bureau of Statistics (NBS). The ARDL Bounds Test confirms a statistically significant long-run cointegrating relationship (F-statistic = 4.502, exceeding all upper critical bounds). Short-run ARDL estimates reveal that current GPI has a large and significant negative effect on GDP per capita ($\beta = -43.22$, $p = 0.0051$), interpreted as a transitional human capital investment cost—not evidence that gender parity is harmful. Current female unemployment significantly reduces GDP per capita ($\beta = -4.69$, $p = 0.0209$), reflecting the immediate cost of excluding women from the labour market. The model explains 53.5% of variance in GDP per capita and passes diagnostic tests for heteroskedasticity and serial correlation. The study recommends strict enforcement of gender equality policies, expanded financial support for girls' education, investment in gender-sensitive school infrastructure, STEM-focused curricula reforms, and labour market reforms to convert female educational gains into sustained economic participation.

Keywords: *Gender inequality, education, economic development, ARDL model, human capital, Nigeria, GDP per capita*

1. INTRODUCTION

Economic development is a multifaceted concept encompassing improvements in GDP, income distribution, poverty reduction, employment, and quality of life (Todaro & Smith, 2012). One of its most consequential determinants is the level of human capital embedded in a nation's population—and education is the primary vehicle through which human capital is accumulated (Schultz, 1961; Becker, 1993). When access to education is systematically unequal across gender lines, the economic costs extend far beyond the individuals affected: they compromise national productivity, innovation capacity, and long-run growth.

Nigeria presents a critical case study. The Federal Republic of Nigeria's National Policy on Education (FRN, 2004) affirms that every Nigerian child shall have equal educational rights irrespective of real or imagined disabilities. Yet in practice, persistent cultural norms—particularly in northern Nigeria—relegate women to domestic roles, contributing to low female enrollment, high dropout rates, and early marriage (Omoregie & Ihensekhien, 2009). UNICEF estimates that nearly 10.5 million Nigerian children are currently out of school, with girls constituting a disproportionate share. The consequence is a large and structurally underutilized female labour force that limits Nigeria's aggregate productive capacity.

While a growing empirical literature documents the relationship between gender inequality in education and economic growth globally (Klasen & Lamanna, 2009; Onogwu, 2021; Adeleke et al., 2024), Nigerian-specific

longitudinal evidence using modern time-series econometrics remains relatively sparse. This study addresses that gap by employing the Autoregressive Distributed Lag (ARDL) framework over a 43-year period (1981–2024) to estimate both the short-run and long-run dynamics of the relationship between gender educational parity and GDP per capita in Nigeria.

1.1 Objectives of the Study

The primary objective is to investigate the impact of gender inequality in education on economic development in Nigeria. Specific objectives are to:

- Analyze how gender inequality in education influences labor market participation, income distribution, and overall economic growth.
- Evaluate the effectiveness of existing policies and interventions aimed at promoting gender equality in education.
- Propose actionable recommendations for policymakers to enhance gender equality in education and maximize its economic benefits.

1.2 Research Hypotheses

- H_1: Gender inequality in education negatively influences economic development in Nigeria.
- H_2: Policies and interventions promoting gender equality in education have a significant positive impact on girls' educational attainment.
- H_3: Gender inequality in education has no significant influence on economic development in Nigeria (null hypothesis).

2. LITERATURE REVIEW

2.1 Conceptual Framework

2.1.1 Gender Inequality and Economic Development

Gender inequality refers to unequal access to resources, opportunities, and privileges based on gender (Onogwu, 2021; Sen, 2001). In educational contexts, it manifests as disparities in enrollment, retention, quality of instruction received, and eventual labor market outcomes. Structural barriers include cultural norms prioritizing boys' education, early marriage practices, lack of gender-sensitive school facilities, and economic constraints on households. These barriers are not merely social—they have measurable, well-documented macroeconomic costs (Klasen & Lamanna, 2009; Wang et al., 2023).

Economic development, as synthesized from Sen (1999), Todaro and Smith (2012), and Schumpeter (1934), is a multidimensional process encompassing not only GDP growth but also structural transformation, human capability expansion, poverty reduction, and the improvement of living standards. Gender inequality in education impedes economic development through at least five channels: reduced human capital stock, lower labor force participation, intergenerational poverty transmission, diminished innovation potential, and inferior health and demographic outcomes.

2.1.2 Addressing Gender Inequality in Education

A comprehensive policy response requires simultaneous action on multiple fronts: enforcing compulsory education legislation with accountability mechanisms at community level; providing conditional cash transfers and scholarships to reduce household economic barriers; improving school infrastructure (particularly gender-segregated sanitation facilities and safe transportation); and reforming curricula to challenge gender stereotypes

in STEM and leadership domains (UNESCO, 2025; UNICEF, 2018). Critically, education policy alone is insufficient if women subsequently face labour market discrimination that prevents their education from translating into economic participation.

2.2 Theoretical Framework

2.2.1 Human Capital Theory

Developed by Schultz (1961) and Becker (1993), human capital theory posits that investments in education and training raise individual productivity, wages, and economic output. Gender inequality in education disrupts this mechanism by creating systematic underinvestment in female human capital. The result is occupational segregation, lower female wages, and a suboptimal allocation of productive capacity. Furthermore, the intergenerational dimension of human capital theory is critical: educated mothers are more likely to invest in their children's schooling, so current gender disparities generate future ones.

2.2.2 Endogenous Growth Theory

Romer (1990) and Lucas (1988) locate the engine of long-run economic growth in the internal accumulation of human capital and the generation of knowledge spillovers. Gender inequality reduces the breadth of this accumulation—excluding women from education limits the diversity of ideas and innovations that drive productivity growth. Nigeria's exclusion of women from STEM and higher education weakens the knowledge economy feedback loops that endogenous growth theory identifies as central to sustained development.

2.2.3 Feminist Economics and Institutional Theory

Feminist economics (Folbre, 1994; Sen, 1999) highlights that conventional economic models undervalue unpaid domestic labor—predominantly female—and that gender biases are embedded in institutional structures rather than merely reflecting market outcomes. Institutional theory (Acemoglu & Robinson, 2012) complements this by noting that both formal institutions (education laws, labor policies) and informal institutions (cultural norms, community expectations) perpetuate educational gender gaps that cannot be corrected by market forces alone.

2.3 Empirical Review

The empirical literature consistently documents a negative relationship between gender educational inequality and economic growth. Klasen and Lamanna (2009), analyzing 139 countries, found that gender gaps in education and employment significantly reduce economic growth, with particularly large effects in Sub-Saharan Africa. Onogwu (2021), using panel data for 29 Sub-Saharan African countries (1996–2019), found gender inequality significantly negatively impacts GDP per capita. Wang et al. (2023) confirmed that gender parity in education has a positive effect on economic growth in 17 sub-Saharan African countries.

For Nigeria specifically, Adeleke et al. (2024) applied ARDL methods to 1991–2022 data and found gender inequality in education had a significant negative long-run impact but a positive short-run relationship with economic growth—consistent with a transitional investment cost interpretation. Lawanson and Umar (2019) using ARDL over 1980–2018 established that gender inequality in both education and employment poses grave consequences for inclusive growth in Nigeria. Nnoje (2024) found a bi-directional Granger-causality relationship between gender inequality and economic growth. These findings collectively motivate the present analysis, which extends the sample through 2024 and employs comprehensive model diagnostics.

3. METHODOLOGY

3.1 Research Design and Data

The study employs a descriptive research design using secondary time-series data spanning 43 years (1981–2024). Data on GDP per capita growth (PC), Gender Parity Index (GPI), Labour Force Participation (LFP), Female Literacy Rate (FLR), Male Literacy Rate (MLR), Government Expenditure on Education (GEE), and Female Unemployment (UEMF) were sourced from the CBN Statistical Bulletin (2024), World Development Indicators (World Bank, 2025), and National Bureau of Statistics (2025).

3.2 Model Specification

The regression model takes the form:

$$PCt = \beta_0 + \beta_1 GPIt + \beta_2 LFPt + \beta_3 FLRt + \beta_4 MLRt + \beta_5 GEEt + \beta_6 UEMFt + \varepsilon$$

Where PC = GDP per capita growth rate; GPI = School Enrollment Gender Parity Index (primary and secondary); LFP = Female-to-male labour force participation ratio; FLR = Female Literacy Rate; MLR = Male Literacy Rate; GEE = Government Expenditure on Education (% GDP); UEMF = Female Unemployment Rate; ε = error term.

3.3 Estimation Strategy

The Augmented Dickey-Fuller (ADF) unit root test was first applied to establish the order of integration for each series. Given the resultant mix of I(0) and I(1) variables, the ARDL bounds testing approach was selected—an established and flexible cointegration methodology that does not require all variables to be integrated of the same order (Pesaran et al., 2001). The ARDL Bounds Test was used to confirm the existence of a long-run cointegrating relationship. Breusch-Pagan-Godfrey and Breusch-Godfrey diagnostic tests were then applied to verify the absence of heteroskedasticity and serial correlation, respectively. All estimations were conducted using EViews, with lag selection based on the Akaike Information Criterion (AIC).

4. RESULTS AND ANALYSIS

4.1 Unit Root Test Results

ADF tests were conducted with constant and trend specifications. Results are summarized in Table 1.

Variable	ADF at Level (prob.)	ADF at 1st Diff. (prob.)	Order of Integration
PC (GDP per capita)	0.0203*	—	I(0)
GPI (Gender Parity Index)	0.0680	0.0000***	I(1)
LFP (Labour Force Participation)	0.0002***	0.0000***	I(1)
FLR (Female Literacy Rate)	0.0051**	—	I(0)
MLR (Male Literacy Rate)	0.8615	0.0001***	I(1)
GEE (Govt. Expenditure on Education)	0.8874	0.0026**	I(1)
UEMF (Female Unemployment)	0.049*	—	I(0)

Table 1: ADF Unit Root Test Results (p<0.10; ** p<0.05; *** p<0.01)*

The mixed integration orders—with PC, FLR, and UEMF stationary at level I(0) and GPI, LFP, MLR, and GEE requiring first differencing I(1)—confirm that ARDL is the appropriate estimation framework.

4.2 ARDL Bounds Test for Cointegration

The Bounds Test examines the null hypothesis that no long-run relationship exists between the variables.

F-Statistic	10% I(0)/I(1)	5% I(0)/I(1)	2.5% I(0)/I(1)	1% I(0)/I(1)	Decision
4.502	2.12 / 3.23	2.45 / 3.61	2.75 / 3.99	3.15 / 4.43	Reject H_0 — Cointegration Exists

Table 2: ARDL Bounds Test Results (k = 6, n = 41)

The F-statistic of 4.502 exceeds the upper bound I(1) at all significance levels (1%, 2.5%, 5%, and 10%), confirming a statistically significant long-run cointegrating relationship among the variables. This validates the ARDL model for estimating both short-run dynamics and long-run equilibrium.

4.3 Diagnostic Tests

Test	Statistic	p-value	Conclusion
Breusch-Pagan-Godfrey (Heteroskedasticity)	F = 1.430	0.2125	No heteroskedasticity
Breusch-Godfrey (Serial Correlation)	F = 0.160	0.8527	No serial correlation
Durbin-Watson Statistic	2.027	—	No autocorrelation

Table 3: Model Diagnostic Test Results

Both diagnostic tests confirm that the model satisfies key OLS assumptions: residuals are homoskedastic and not serially correlated, ensuring efficient and reliable coefficient estimates.

4.4 ARDL Estimation Results

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PC(-1)	0.3997	0.1289	3.099	0.0043**
GPI (current)	−43.2245	14.272	−3.029	0.0051**
GPI(-1)	24.4668	13.184	1.856	0.0737†
LFP	57.4782	66.450	0.865	0.3941
FLR	0.1110	0.268	0.415	0.6813
MLR	−0.3177	0.430	−0.739	0.4657
GEE	−0.1536	0.762	−0.202	0.8417
UEMF (current)	−4.6889	1.920	−2.442	0.0209*
UEMF(-1)	9.6026	3.185	3.015	0.0053**

Variable	Coefficient	Std. Error	t-Statistic	Prob.
UEMF(-2)	−11.1446	4.141	−2.691	0.0117*
UEMF(-3)	7.2151	2.430	2.969	0.0059**
Constant	−19.2175	66.184	−0.290	0.7736

Table 4: ARDL Short-Run Estimates (Dependent variable: PC; $R^2 = 0.535$, Adj. $R^2 = 0.358$, $F = 3.029$, $p = 0.0082$)

*† p<0.10; * p<0.05; ** p<0.01*

4.5 Interpretation of Results

GDP Persistence (PC(-1) = 0.40, p = 0.004): The lagged dependent variable is positive and highly significant. A 1% increase in last year's GDP per capita produces a 0.40% increase this year, confirming strong economic inertia—path dependence consistent with structural investment patterns, population effects, and production capacity persistence.

Gender Parity Index — Short-Run Cost (GPI = −43.22, p = 0.005): The large negative and highly significant coefficient on current GPI is the central finding of this study. It indicates that short-run improvements in school enrollment gender parity are associated with a contemporaneous decline in GDP per capita. This is interpreted as a transitional human capital investment cost rather than evidence that gender equality is economically harmful. When girls enter school—especially in rural or low-income communities—they are redirected from informal productive activities (domestic labor, market trading, subsistence farming). Household income temporarily declines. Moreover, students in school are not yet economically productive; education generates returns only after a lag of several years. The lagged GPI coefficient (24.47, p = 0.074) partially reverting towards positive territory is consistent with this interpretation. Long-run gains from gender parity—as confirmed by the Bounds Test cointegration—will eventually dominate.

Labour Force Participation (LFP = 57.48, p = 0.394): The large positive coefficient—though statistically insignificant—suggests that female labour force participation contributes positively to GDP per capita in direction, but that the quality of employment matters more than participation per se. In Nigeria's economy, where underemployment and informal sector work are pervasive, raw participation rates do not translate reliably to productivity gains.

Literacy Rates (FLR = 0.11, p = 0.681; MLR = −0.32, p = 0.466): Both literacy variables are statistically insignificant. Female literacy has the expected positive sign but has not yet achieved the critical mass needed for statistical significance, likely due to the historically low levels of female literacy in Nigeria over the sample period. Male literacy shows a negative (though insignificant) coefficient, possibly reflecting diminishing marginal returns or a misalignment between educational content and labor market needs.

Government Education Expenditure (GEE = −0.15, p = 0.842): The negative and highly insignificant coefficient is concerning. It suggests that public expenditure on education in Nigeria is not generating commensurate economic returns—possibly due to fund misallocation, corruption, or poor policy design. Spending without targeted outcomes produces no growth dividend.

Female Unemployment Dynamics (UEMF lags): Current female unemployment reduces GDP per capita significantly (−4.69, p = 0.021), confirming the immediate cost of excluding women from productive employment. The oscillating lagged effects (UEMF(-1) = +9.60, UEMF(-2) = −11.14, UEMF(-3) = +7.22, all significant) reflect the cyclical and structurally unstable nature of female labour market integration in Nigeria—women enter and exit the labor market in response to external shocks, household constraints, and policy changes, producing volatile economic contributions.

5. DISCUSSION

5.1 The Short-Run Cost / Long-Run Gain Paradox

The central empirical finding—that GPI improvements are negatively associated with contemporaneous GDP per capita—is theoretically coherent and not unprecedented in the literature. Adeleke et al. (2024) similarly found a negative short-run but positive long-run relationship between gender inequality in education and economic growth in Nigeria using ARDL. The ARDL Bounds Test result in the present study independently confirms the existence of a long-run cointegrating relationship, meaning that the negative short-run effect is part of a transitional adjustment process toward a higher long-run equilibrium. This parallels the logic of any human capital investment: near-term resource costs precede productivity gains.

The policy implication is important: the negative short-run GDP coefficient should not be used to argue against gender parity investments. On the contrary, it underscores why sustained, multi-year commitment to girls' education is necessary. Governments and international organizations must be prepared to absorb transitional costs and maintain political will through the lag period before returns materialize.

5.2 Labour Market Barriers and the Education-Employment Gap

The insignificance of labour force participation and the oscillating female unemployment dynamics jointly point to a structural mismatch: Nigeria is not efficiently converting female educational gains into labour market outcomes. Even as GPI has trended upward from 0.75 in 1981 to 0.99 in 2024, female unemployment has remained persistently high and volatile. This is consistent with Enyioko (2021), who documented persistent gender gaps in university enrollment and academic representation, and with Egbulonu and Eleonu (2018), who found that the female employment level has increased without a commensurate reduction in discrimination.

The implication is that education-side interventions, while necessary, are not sufficient. Labour market reforms—including equal opportunity enforcement, entrepreneurship grants, and childcare support—are required to translate female educational gains into sustained economic participation.

5.3 Efficiency of Public Education Spending

The near-zero and negative coefficient on Government Expenditure on Education corroborates a consistent finding in the Nigerian empirical literature: public education spending is not generating adequate returns. Lawanson and Umar (2019) similarly found that gender inequality in education portends grave consequences for inclusive growth despite policy pronouncements. The National Policy on Inclusive Education exists; implementation and accountability mechanisms do not. Reforming education spending toward measurable outcomes—girls' enrollment, retention, and labor market placement—is essential.

6. CONCLUSION AND RECOMMENDATIONS

6.1 Conclusion

This study provides robust empirical evidence that gender inequality in education negatively impacts Nigeria's long-run economic development. The ARDL Bounds Test confirms a significant long-run cointegrating relationship among the study variables. The short-run negative effect of GPI on GDP per capita is interpreted as a transitional investment cost—analogous to planting a tree that will bear fruit only after several years of cultivation—and is consistent with prior Nigerian evidence. The cyclical instability of female unemployment further demonstrates that Nigeria's labour market is failing to absorb educated women efficiently, a structural failure that must be addressed alongside education-side policies.

Nigeria cannot achieve Sustainable Development Goal 4 or SDG 8 without making equitable female education and female economic participation national developmental priorities. Every girl denied education is not only

denied her individual potential—she represents a quantifiable reduction in the nation's productive capacity, innovation ecosystem, and long-run GDP trajectory.

6.2 Recommendations

- Strengthen Policy Implementation: The Universal Basic Education program and National Gender Policy must be rigorously enforced, particularly in rural and northern regions. Community-based accountability mechanisms and penalties for discriminatory practices such as early marriage are essential.
- Expand Financial Support for Female Education: Scholarship programs, conditional cash transfers, and school feeding initiatives targeting low-income families should be scaled. Public-private partnerships should fund vocational training and STEM programs for girls.
- Invest in Gender-Sensitive School Infrastructure: Gender-segregated sanitation facilities, safe transportation, anti-harassment policies, and female teacher recruitment improve enrollment and retention. Digital learning platforms can extend reach to remote areas.
- Reform Curricula and Teacher Training: Educational materials should be audited for gender stereotypes. STEM and leadership curricula should be explicitly designed to include girls. Mentorship programs featuring successful female professionals should be institutionalized.
- Address Labour Market Barriers: Equal opportunity law enforcement, entrepreneurship grants, flexible work policies, and childcare support are necessary complements to education investments. Industry collaboration to eliminate hiring biases in high-growth sectors is critical.
- Launch Nationwide Awareness Campaigns: Media campaigns, community dialogues, and religious leader engagement should communicate the economic returns to educating girls, challenging norms that prioritize boys' education.
- Reform Education Spending Accountability: The negative and insignificant GEE coefficient signals resource misallocation. Budget allocations must be tied to measurable outcomes—enrollment, completion, and employment rates disaggregated by gender—with independent monitoring and public reporting.
- Enhance Data Infrastructure: A robust system tracking gender disparities in enrollment, completion, and employment outcomes will enable iterative, evidence-based policy improvement and support future longitudinal research.

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